

Lecture 19

Cost Analysis of the Firm

ECO303

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Definitions of Costs

- It is important to differentiate between accounting cost and economic cost
 - Accountants: out-of-pocket expenses, historical costs, depreciation, and other bookkeeping entries
 - Economists focus more on opportunity cost

Definitions of Costs

- Labor Costs

- Accountants: expenditures on labor are current expenses
- Economists: labor is an explicit cost
 - Labor services are contracted at an hourly wage (w)
 - It is assumed that this is also what the labor could earn in alternative employment

Definitions of Costs

- Capital Costs

- Accountants: use the historical price of the capital and apply some depreciation rule to determine current costs
- Economists: refer to the capital's original price as a “sunk cost”
 - Implicit cost of the capital - what someone else would be willing to pay for its use
 - We will use v to denote the rental rate for capital

Definitions of Costs

- **Costs of Entrepreneurial Services**
 - Accountants: the owner of a firm is entitled to all profits
 - Revenues or losses left over after paying all input costs
 - Economists: the opportunity costs of time and funds that owners devote to the operation of their firms
 - Part of accounting profits would be considered as entrepreneurial costs by economists

Economic Cost

- Economic cost of any input
 - The payment required to keep that input in its present employment
 - The remuneration the input would receive in its best alternative employment

Simplifying Assumptions

- There are only two inputs
 - Homogeneous labor (l), measured in labor-hours
 - Homogeneous capital (k), measured in machine-hours
 - Entrepreneurial costs - included in capital costs
- Inputs are hired in perfectly competitive markets
 - Firms are price takers in input markets

Economic Profits

- Total costs for the firm:

$$\text{total costs} = C = wl + vk$$

- Total revenue for the firm:

$$\text{total revenue} = pq = pf(k,l)$$

- Economic profits (π):

$$\pi = \text{total revenue} - \text{total cost}$$

$$\pi = pq - wl - vk$$

$$\pi = pf(k,l) - wl - vk$$

Economic Profits

- Economic profits
 - Are a function of the amount of k and l employed
 - We could examine how a firm would choose k and l to maximize profit
 - “Derived demand” theory of labor and capital inputs
 - Assume that the firm has already chosen its output level (q_0) and wants to minimize its costs

Cost-Minimizing Input Choices

- Minimum cost
 - Occurs where the $RTS = w/v$
 - The rate at which k can be traded for l in the production process = the rate at which they can be traded in the marketplace

Cost-Minimizing Input Choices

- Minimize total costs given $q = f(k,l) = q_0$
- Setting up the Lagrangian:

$$\mathcal{L} = wl + vk + \lambda[q_0 - f(k,l)]$$

- First-order conditions:

$$\partial \mathcal{L} / \partial l = w - \lambda(\partial f / \partial l) = 0$$

$$\partial \mathcal{L} / \partial k = v - \lambda(\partial f / \partial k) = 0$$

$$\partial \mathcal{L} / \partial \lambda = q_0 - f(k,l) = 0$$

Cost-Minimizing Input Choices

- Dividing the first two conditions we get

$$\frac{w}{v} = \frac{\partial f / \partial l}{\partial f / \partial k} = RTS \text{ (} l \text{ for } k \text{)}$$

- The cost-minimizing firm should equate the RTS for the two inputs to the ratio of their prices

Cost-Minimizing Input Choices

- Cross-multiplying, we get

$$\frac{f_k}{v} = \frac{f_l}{w}$$

- For costs to be minimized, the marginal productivity per dollar spent should be the same for all inputs

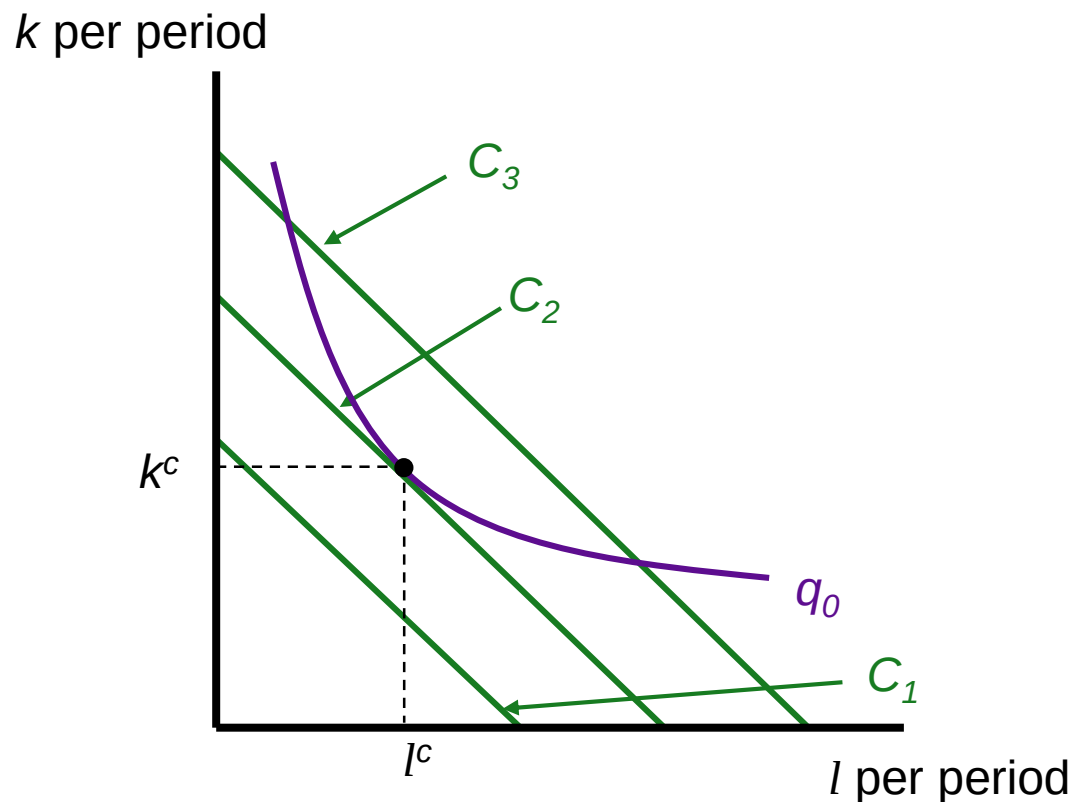
Cost-Minimizing Input Choices

- The inverse of this equation is also of interest

$$\frac{w}{f_l} = \frac{v}{f_k} = \lambda$$

- The Lagrangian multiplier shows how the extra costs that would be incurred by increasing the output constraint slightly

FIGURE 10.1

Minimization of Costs Given $q = q_0$ 

A firm is assumed to choose k and l to minimize total costs. The condition for this minimization is that the rate at which k and l can be traded technically (while keeping $q = q_0$) should be equal to the rate at which these inputs can be traded in the market. In other words, the *RTS* (of l for k) should be set equal to the price ratio w/v . This tangency is shown in the figure; costs are minimized at C_1 by choosing inputs k^c and l^c .

Contingent Demand for Inputs

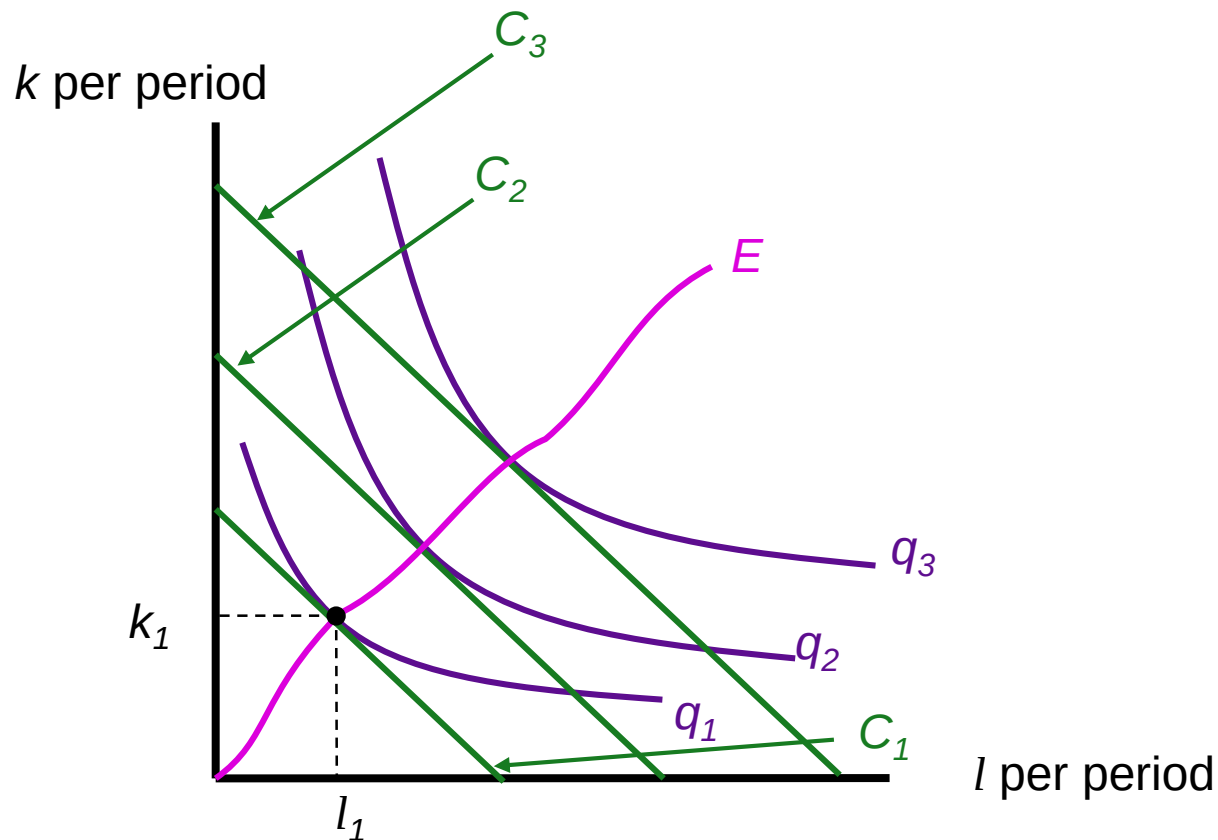
- Cost minimization
 - Leads to a demand for capital and labor that is contingent on the level of output being produced
- The demand for an input is a derived demand
 - It is based on the level of the firm's output

The Firm's Expansion Path

- The firm can determine
 - The cost-minimizing combinations of k and l for every level of output
- If input costs remain constant for all amounts of k and l
 - We can trace the locus of cost-minimizing choices
 - Called the firm's expansion path

FIGURE 10.2

The Firm's Expansion Path



The firm's expansion path is the locus of cost-minimizing tangencies. Assuming fixed input prices, the curve shows how inputs increase as output increases.

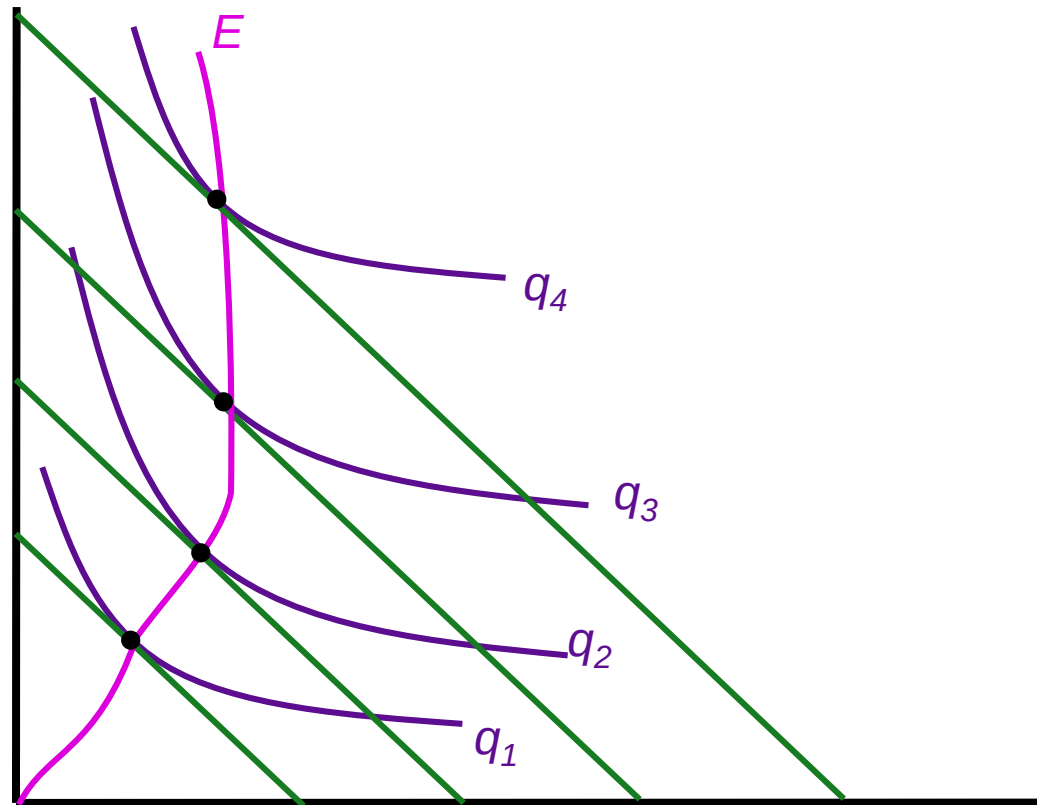
The Firm's Expansion Path

- The expansion path does not have to be a straight line
 - The use of some inputs may increase faster than others as output expands
 - Depends on the shape of the isoquants
- The expansion path does not have to be upward sloping
 - If the use of an input falls as output expands, that input is an inferior input

FIGURE 10.3

Input Inferiority

k per period



l per period

With this particular set of isoquants, labor is an inferior input because less l is chosen as output expands beyond q_2 .

EXAMPLE 10.1 Cost Minimization

- Cobb-Douglas production function: $q = k^\alpha l^\beta$
 - The Lagrangian expression for cost minimization of producing q_0 is

$$\mathcal{L} = vk + wl + \lambda(q_0 - k^\alpha l^\beta)$$

- First-order conditions for a minimum

$$\partial \mathcal{L} / \partial k = v - \lambda \alpha k^{\alpha-1} l^\beta = 0$$

$$\partial \mathcal{L} / \partial l = w - \lambda \beta k^\alpha l^{\beta-1} = 0$$

$$\partial \mathcal{L} / \partial \lambda = q_0 - k^\alpha l^\beta = 0$$

EXAMPLE 10.1 Cost Minimization

- Dividing the first equation by the second gives us

$$\frac{w}{v} = \frac{\beta k^{\alpha} l^{\beta-1}}{\alpha k^{\alpha-1} l^{\beta}} = \frac{\beta}{\alpha} \cdot \frac{k}{l} = RTS$$

- This production function is homothetic
 - The RTS depends only on the ratio of the two inputs
 - The expansion path is a straight line

EXAMPLE 10.1 Cost Minimization

- CES production function: $q = (k^\rho + l^\rho)^{\gamma/\rho}$
 - The Lagrangian expression for cost minimization of producing q_0 is

$$\mathcal{L} = vk + wl + \lambda[q_0 - (k^\rho + l^\rho)^{\gamma/\rho}]$$

- First-order conditions for a minimum

$$\partial \mathcal{L} / \partial k = v - \lambda(\gamma/\rho)(k^\rho + l^\rho)^{(\gamma-\rho)/\rho}(\rho)k^{\rho-1} = 0$$

$$\partial \mathcal{L} / \partial l = w - \lambda(\gamma/\rho)(k^\rho + l^\rho)^{(\gamma-\rho)/\rho}(\rho)l^{\rho-1} = 0$$

$$\partial \mathcal{L} / \partial \lambda = q_0 - (k^\rho + l^\rho)^{\gamma/\rho} = 0$$

EXAMPLE 10.1 Cost Minimization

- Dividing the first equation by the second gives us

$$\frac{w}{v} = \left(\frac{1}{k} \right)^{\rho-1} = \left(\frac{k}{l} \right)^{1-\rho} = \left(\frac{k}{l} \right)^{1/\sigma}$$

- This production function is also homothetic

Total Cost Function

- Total cost function

- Shows that for any set of input costs and for any output level
- The minimum cost incurred by the firm is

$$C = C(v, w, q)$$

- As output (q) increases, total costs increase

Cost Functions

- Average cost function (AC)

- Is found by computing total costs per unit of output

$$\text{average cost} = AC(v, w, q) = \frac{C(v, w, q)}{q}$$

- Marginal cost function (MC)

- Is found by computing the change in total costs for a change in output produced

$$\text{marginal cost} = MC(v, w, q) = \frac{\partial C(v, w, q)}{\partial q}$$

Graphical Analysis of Total Costs

- To produce one unit of output we need
 - k_1 units of capital
 - And l_1 units of labor input

$$C(q=1) = vk_1 + wl_1$$

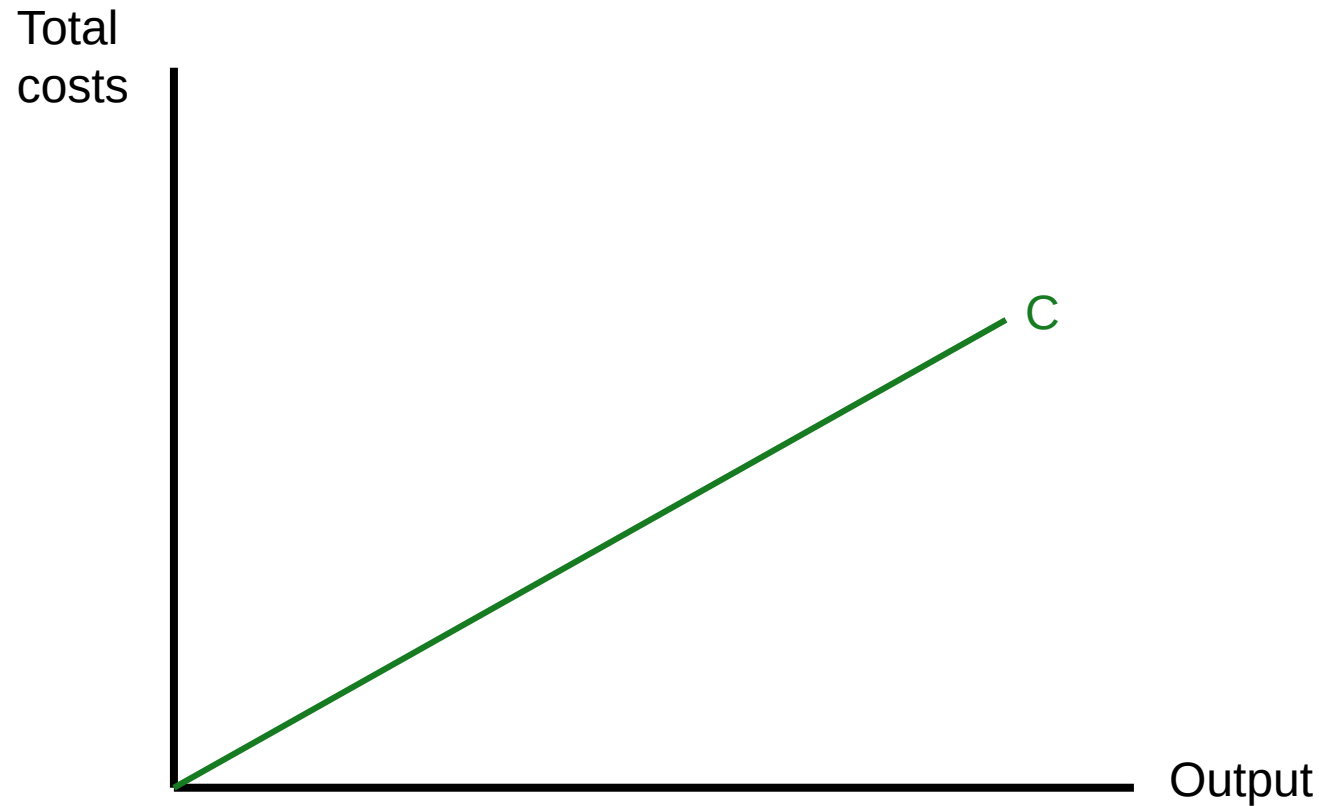
- To produce m units of output
 - Assuming constant returns to scale

$$C(q=m) = vmk_1 + wml_1 = m(vk_1 + wl_1)$$

$$C(q=m) = m \cdot C(q=1)$$

FIGURE 10.4 (a)

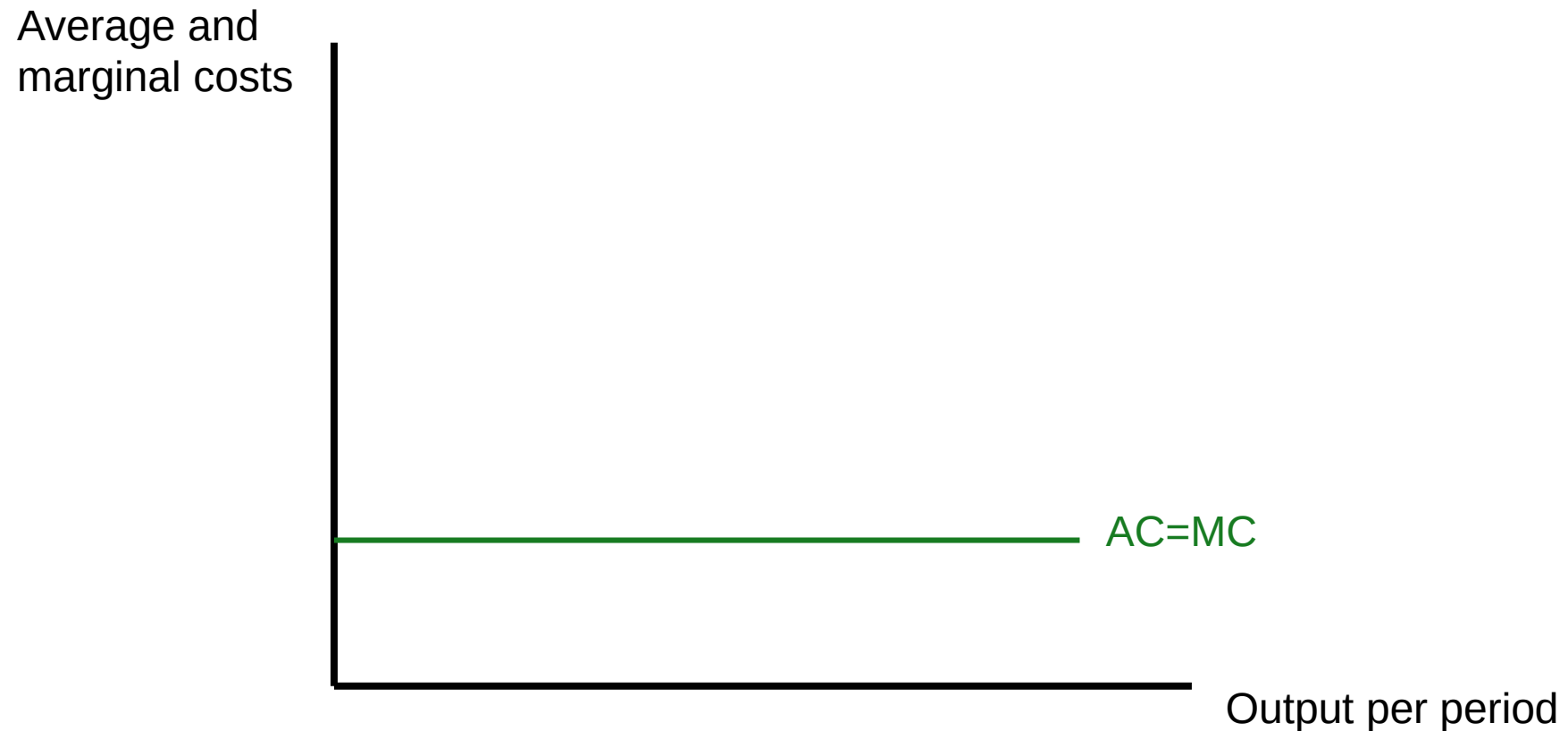
Total, Average, and Marginal Cost Curves for the Constant Returns-to-Scale Case



In (a) total costs are proportional to output level.

FIGURE 10.4 (b)

Total, Average, and Marginal Cost Curves for the Constant Returns-to-Scale Case



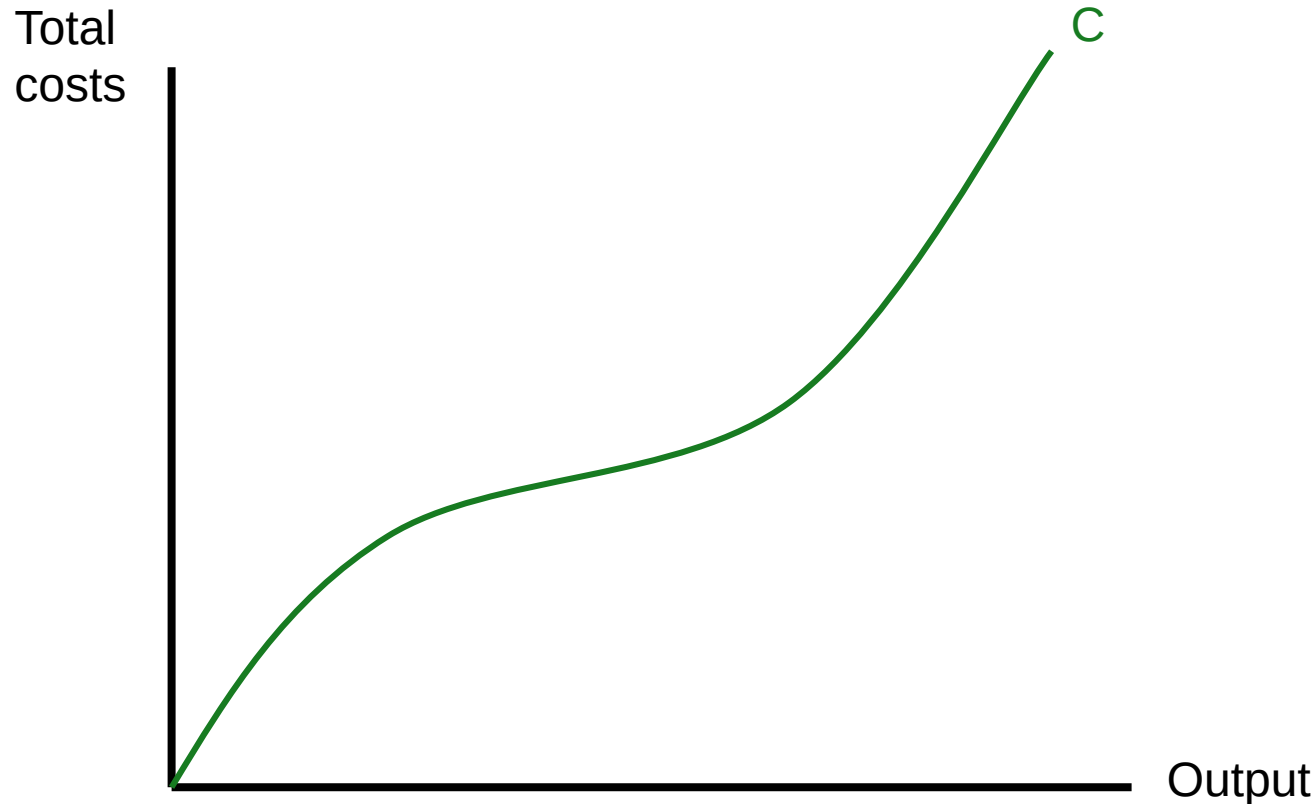
Average and marginal costs, as shown in (b), are equal and constant for all output levels.

Graphical Analysis of Total Costs

- Suppose that total costs start out as concave and then becomes convex as output increases
 - One possible explanation for this is that there is a third factor of production that is fixed as capital and labor usage expands
 - Total costs begin rising rapidly after diminishing returns set in

FIGURE 10.5 (a)

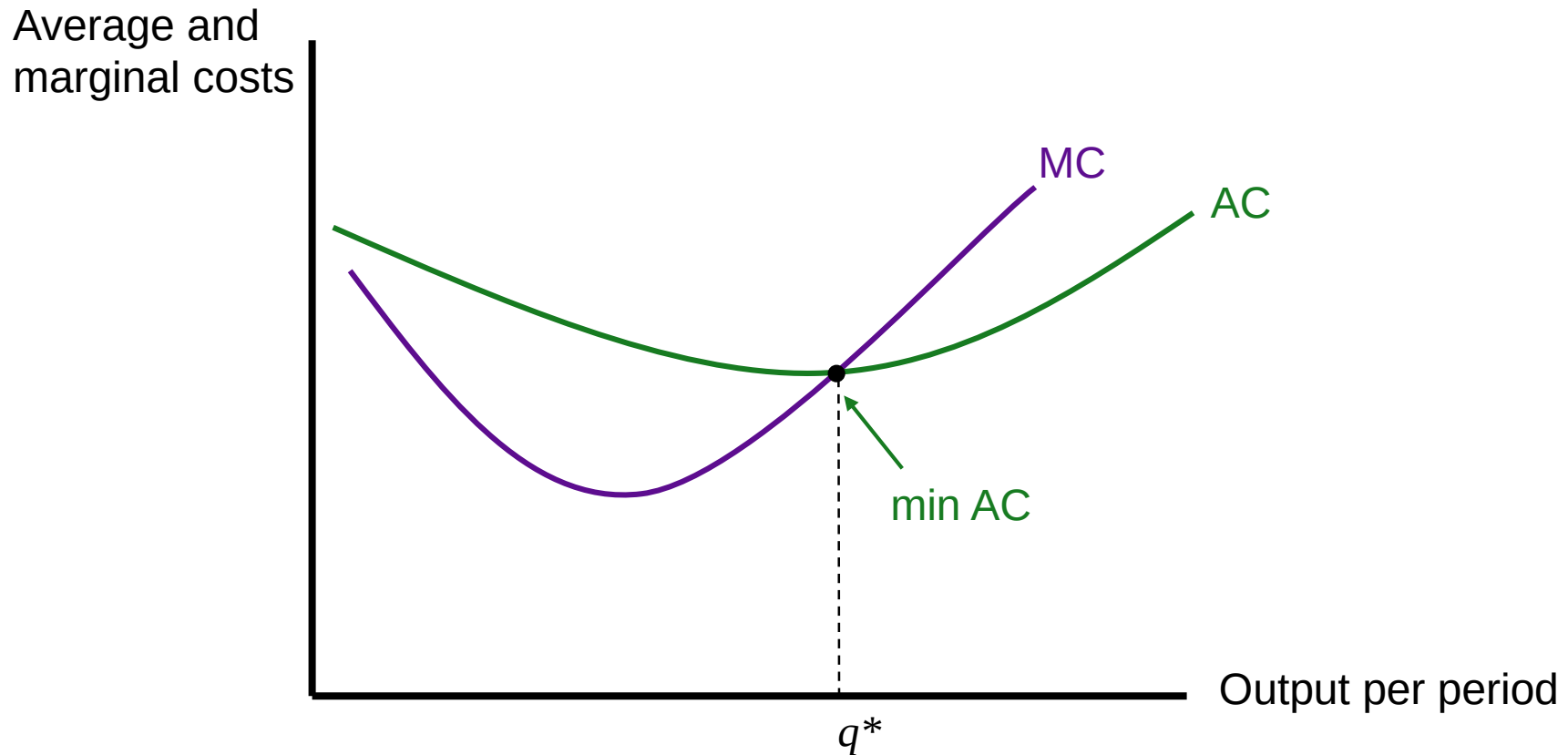
Total, Average, and Marginal Cost Curves for the Cubic Total Cost Curve Case



If the total cost curve has the cubic shape shown in (a), average and marginal cost curves will be U-shaped.

FIGURE 10.5 (b)

Total, Average, and Marginal Cost Curves for the Cubic Total Cost Curve Case



If the total cost curve has the cubic shape shown in (a), average and marginal cost curves will be U-shaped. In (b) the marginal cost curve passes through the low point of the average cost curve at output level q^* .

Shifts in Cost Curves

- Cost curves
 - Are drawn under the assumption that input prices and the level of technology are held constant
 - Any change in these factors will cause the cost curves to shift

- Fixed proportions

$$q = f(k, l) = \min(\alpha k, \beta l)$$

- Production will occur at the vertex of the L-shaped isoquants ($q = ak = bl$)

$$C(w, v, q) = vk + wl = v(q/a) + w(q/b)$$

$$C(w, v, q) = a \left(\frac{v}{a} + \frac{w}{b} \right)$$

EXAMPLE 10.2 Some Illustrative Cost Functions

- Cobb-Douglas, $q = f(k, l) = k^\alpha l^\beta$
 - Cost minimization requires that:

$$\frac{w}{v} = \frac{\beta}{\alpha} \cdot \frac{k}{l}, \text{ so } k = \frac{\alpha}{\beta} \cdot \frac{w}{v} \cdot l$$

- Substitute into the production function and solve for l , then for k

$$l = q^{1/\alpha+\beta} \left(\frac{\beta}{\alpha} \right)^{\alpha/\alpha+\beta} w^{-\alpha/\alpha+\beta} v^{\alpha/\alpha+\beta}$$

$$k = q^{1/\alpha+\beta} \left(\frac{\alpha}{\beta} \right)^{\beta/\alpha+\beta} w^{\beta/\alpha+\beta} v^{-\beta/\alpha+\beta}$$

- Cobb-Douglas

- Now we can derive total costs as

$$C(v, w, q) = vk + wl = q^{1/\alpha+\beta} B v^{\alpha/\alpha+\beta} w^{\beta/\alpha+\beta}$$

- Where

$$B = (\alpha + \beta) \alpha^{-\alpha/\alpha+\beta} \beta^{-\beta/\alpha+\beta}$$

- Which is a constant that involves only the parameters α and β

EXAMPLE 10.2 Some Illustrative Cost Functions

- CES: $q = f(k, l) = (k^\rho + l^\rho)^{1/\rho}$
 - To derive the total cost, we would use the same method and eventually get

$$C(v, w, q) = vk + wl = q^{1/\gamma} (v^{\rho/\rho-1} + w^{\rho/\rho-1})^{(\rho-1)/\rho}$$

$$C(v, w, q) = q^{1/\gamma} (v^{1-\sigma} + w^{1-\sigma})^{1/1-\sigma}$$

Properties of Cost Functions

- Homogeneity

- Cost functions are all homogeneous of degree one in the input prices
- A doubling of all input prices will not change the levels of inputs purchased
- Inflation will shift the cost curves up

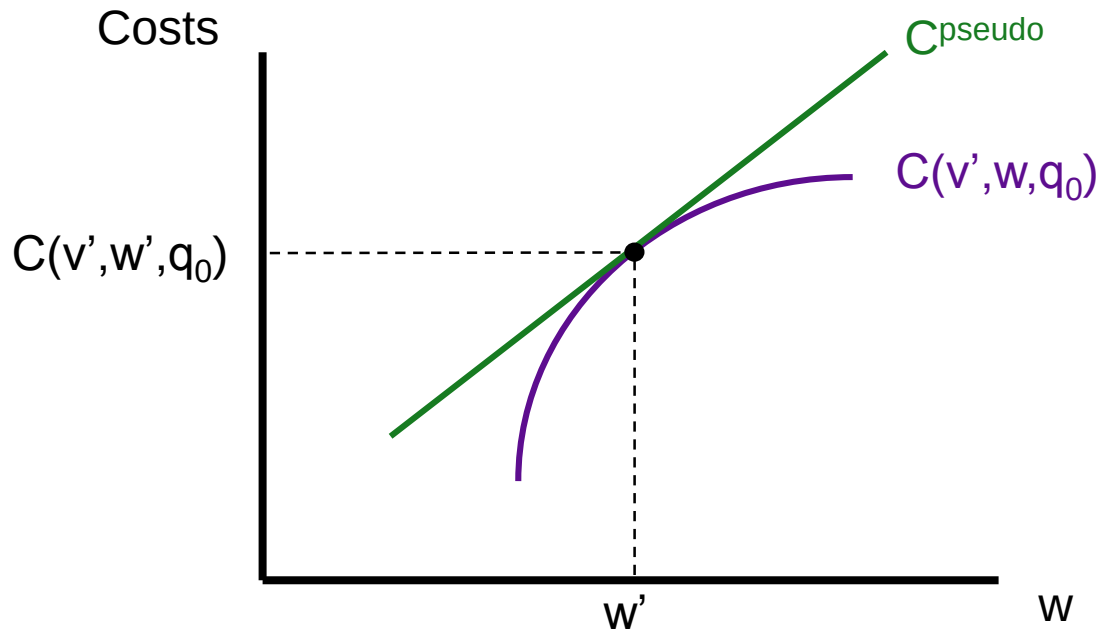
Properties of Cost Functions

- Nondecreasing in q , v , and w
 - Cost functions are derived from a cost-minimization process
 - Any decline in costs from an increase in one of the function's arguments would lead to a contradiction

Properties of Cost Functions

- Concave in input prices
 - Costs will be lower
 - When a firm faces input prices that fluctuate around a given level
 - Than when they remain constant at that level
 - The firm can adapt its input mix to take advantage of such fluctuations

Cost Functions Are Concave in Input Prices



With input prices w' and v' , total costs of producing q_0 are $C(v', w', q_0)$. If the firm does not change its input mix, costs of producing q_0 would follow the straight line C^{PSEUDO} . With input substitution, actual costs $C(v', w, q_0)$ will fall below this line, and hence the cost function is concave in w .

Properties of Cost Functions

- Some of these properties carry over to average and marginal costs
 - Homogeneity
 - Effects of v , w , and q are ambiguous

Input Substitution

- A change in the price of an input
 - Will cause the firm to alter its input mix
 - The change in k/l in response to a change in w/v , while holding q constant is

$$\frac{\partial \left(\frac{k}{l} \right)}{\partial \left(\frac{w}{v} \right)}$$

Input Substitution

- Putting this in proportional terms as

$$s = \frac{d(k/l)}{d(w/v)} \cdot \frac{w/v}{k/l} = \frac{d \ln(k/l)}{d \ln(w/v)}$$

- Gives an alternative definition of the elasticity of substitution
 - In the two-input case, s must be nonnegative
 - Large values of s indicate that firms change their input mix significantly if input prices change

Quantitative Size of Shifts in Costs Curves

- The increase in costs will be largely influenced by
 - The relative significance of the input in the production process
 - The ability of firms to substitute another input for the one that has risen in price

EXAMPLE 10.3 Shifting the Cobb–Douglas Cost Function

- Cobb-Douglas cost function:

$$C(v, w, q) = vk + wl = q^{1/\alpha+\beta} B v^{\alpha/\alpha+\beta} w^{\beta/\alpha+\beta}$$

where $B = (\alpha + \beta) \alpha^{-\alpha/\alpha+\beta} \beta^{-\beta/\alpha+\beta}$

- Assume $\alpha = \beta = 0.5$, the total cost curve is greatly simplified

$$C(v, w, q) = vk + wl = 2qv^{0.5}w^{0.5}$$

EXAMPLE 10.3 Shifting the Cobb–Douglas Cost Function

- If $v = 3$ and $w = 12$, the relationship is

$$C(3, 12, q) = 2q\sqrt{36} = 12q$$

- $C = 480$ to produce $q = 40$
- $AC = C/q = 12$
- $MC = \partial C / \partial q = 12$

- If $v = 3$ and $w = 27$, the relationship is

$$C(3, 27, q) = 2q\sqrt{81} = 18q$$

- $C = 720$ to produce $q = 40$
- $AC = C/q = 18$
- $MC = \partial C / \partial q = 18$

Contingent Demand for Inputs

- Contingent demand functions
 - For all of the firms inputs can be derived from the cost function
- Shephard's lemma
 - The contingent demand function for any input is given by the partial derivative of the total-cost function with respect to that input's price

Contingent Demand for Inputs

- **Shepherd's lemma**
 - Is one result of the envelope theorem
 - The change in the optimal value
 - In a constrained optimization problem
 - With respect to one of the parameters
 - Can be found by differentiating the Lagrangian with respect to the changing parameter

EXAMPLE 10.4 Contingent Input Demand Functions

- Fixed proportions, $C(v, w, q) = q(v/\alpha + w/\beta)$
 - Contingent demand functions are quite simple:

$$k^c(v, w, q) = \frac{\partial C(v, w, q)}{\partial v} = \frac{q}{\alpha}$$

$$l^c(v, w, q) = \frac{\partial C(v, w, q)}{\partial w} = \frac{q}{\beta}$$

EXAMPLE 10.4 Contingent Input Demand Functions

- Cobb-Douglas:

$$C(v, w, q) = vk + wl = q^{1/\alpha+\beta} B v^{\alpha/\alpha+\beta} w^{\beta/\alpha+\beta}$$

- Contingent demand functions:

$$k^c(v, w, q) = \frac{\partial C}{\partial v} = \frac{\alpha}{\alpha + \beta} \cdot q^{1/\alpha+\beta} B v^{-\beta/\alpha+\beta} w^{\beta/\alpha+\beta}$$

$$= \frac{\alpha}{\alpha + \beta} \cdot q^{1/\alpha+\beta} B \left(\frac{w}{v} \right)^{\beta/\alpha+\beta}$$

$$l^c(v, w, q) = \frac{\partial C}{\partial w} = \frac{\beta}{\alpha + \beta} \cdot q^{1/\alpha+\beta} B v^{\alpha/\alpha+\beta} w^{-\alpha/\alpha+\beta}$$

$$= \frac{\beta}{\alpha + \beta} \cdot q^{1/\alpha+\beta} B \left(\frac{w}{v} \right)^{-\alpha/\alpha+\beta}$$

EXAMPLE 10.4 Contingent Input Demand Functions

- CES:

$$C(v, w, q) = q^{1/\gamma} \left(v^{1-\sigma} + w^{1-\sigma} \right)^{\sigma/(1-\sigma)}$$

- The contingent demand functions:

$$k^c(v, w, q) = \frac{\partial C}{\partial v} = \frac{1}{1-\sigma} \cdot q^{1/\gamma} \left(v^{1-\sigma} + w^{1-\sigma} \right)^{\sigma/(1-\sigma)} (1-\sigma) v^{-\sigma}$$

$$= q^{1/\gamma} \left(v^{1-\sigma} + w^{1-\sigma} \right)^{\sigma/(1-\sigma)} v^{-\sigma}$$

$$l^c(v, w, q) = \frac{\partial C}{\partial w} = \frac{1}{1-\sigma} \cdot q^{1/\gamma} \left(v^{1-\sigma} + w^{1-\sigma} \right)^{\sigma/(1-\sigma)} (1-\sigma) w^{-\sigma}$$

$$= q^{1/\gamma} \left(v^{1-\sigma} + w^{1-\sigma} \right)^{\sigma/(1-\sigma)} w^{-\sigma}$$

The Elasticity of Substitution

- Shepherd's lemma

- Can be used to derive information about input substitution directly from the total cost function

$$s_{i,j} = \frac{\partial \ln(x_i/x_j)}{\partial \ln(w_j/w_i)} = \frac{\partial \ln(C_i/C_j)}{\partial \ln(w_j/w_i)}$$

Short-Run, Long-Run Distinction

- In the short run
 - Economic actors have only limited flexibility in their actions
- Assume
 - The capital input is held constant at k_1
 - The firm is free to vary only its labor input
- The production function becomes

$$q = f(k_1, l)$$

Short-Run Total Costs

- Short-run total cost for the firm is

$$SC = vk_1 + wl$$

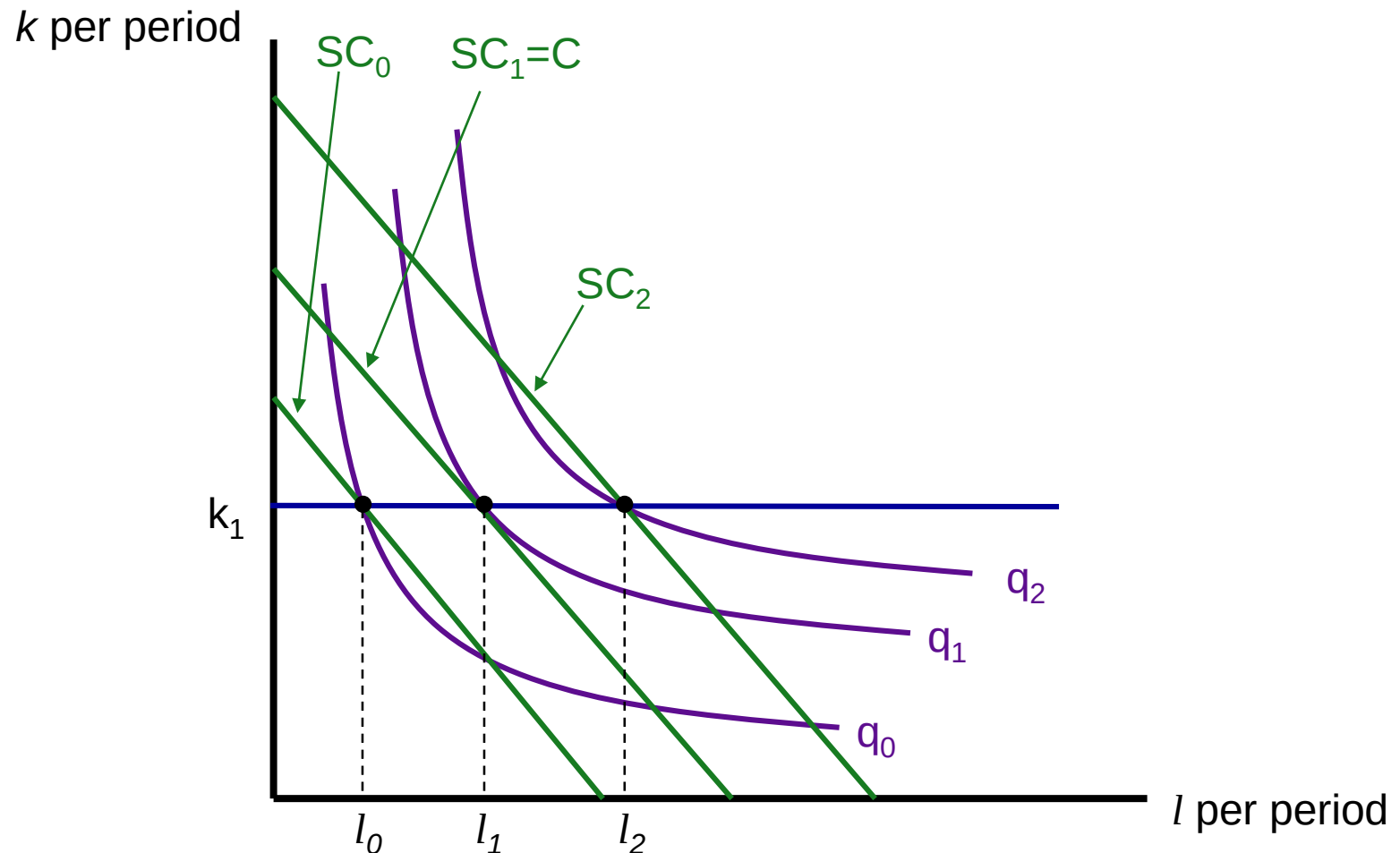
- There are two types of short-run costs:
 - Short-run fixed costs are costs associated with fixed inputs (vk_1)
 - Short-run variable costs are costs associated with variable inputs (wl)

Short-Run Total Costs

- Short-run costs
 - Are not minimal costs for producing the various output levels
 - The firm does not have the flexibility of input choice
 - To vary its output in the short run, the firm must use nonoptimal input combinations
 - The *RTS* will not be equal to the ratio of input prices

FIGURE 10.7

“Nonoptimal” Input Choices Must Be Made in the Short Run



Because capital input is fixed at k , in the short run the firm cannot bring its *RTS* into equality with the ratio of input prices. Given the input prices, q_0 should be produced with more labor and less capital than it will be in the short run, whereas q_2 should be produced with more capital and less labor than it will be.

Short-Run Marginal and Average Costs

- The short-run average total cost (SAC) function is

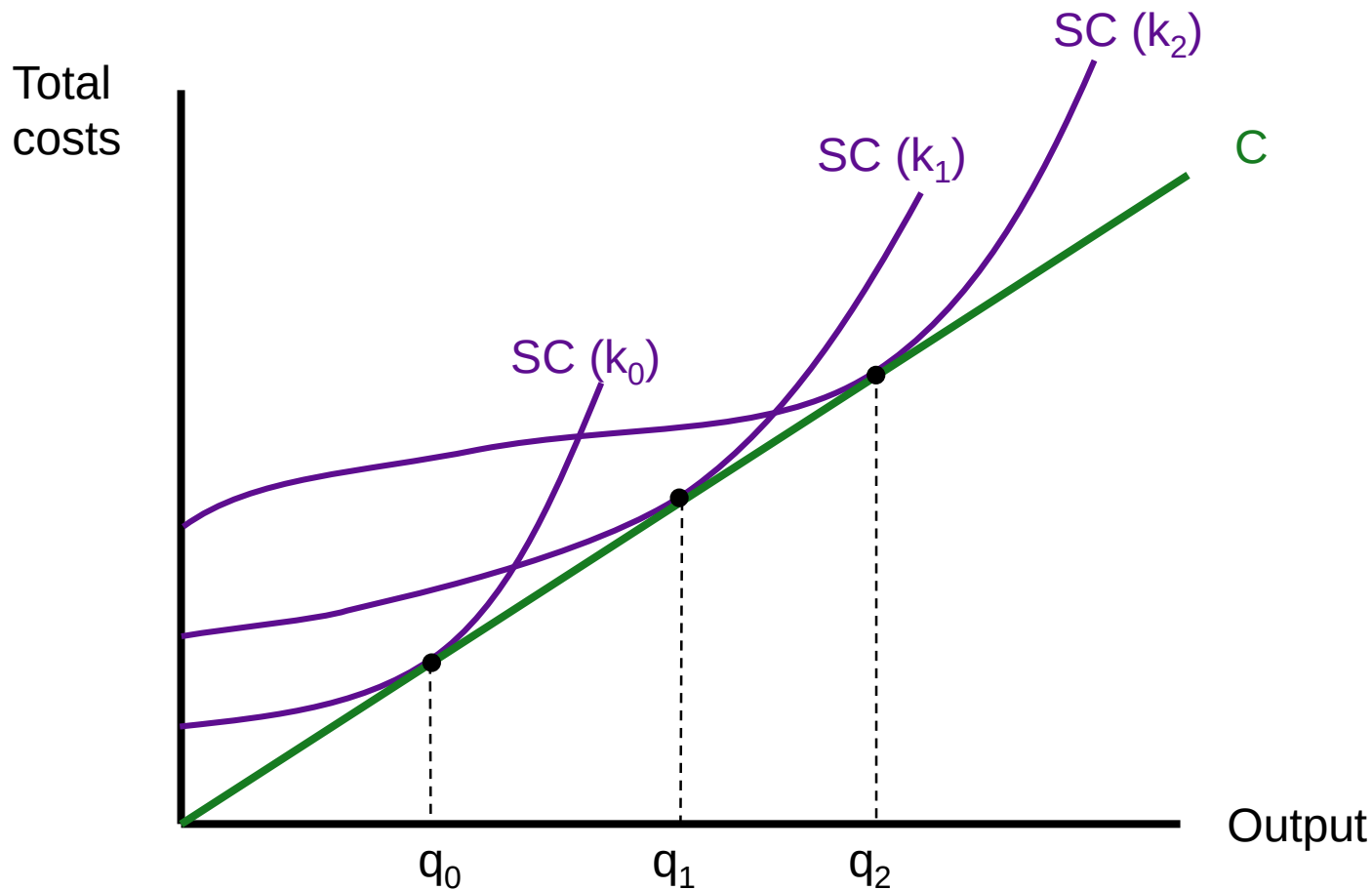
$$SAC = \text{total costs/total output} = SC/q$$

- The short-run marginal cost (SMC) function is

$$SMC = \text{change in } SC/\text{change in output} = \partial SC/\partial q$$

FIGURE 10.8 (a) Constant returns to scale

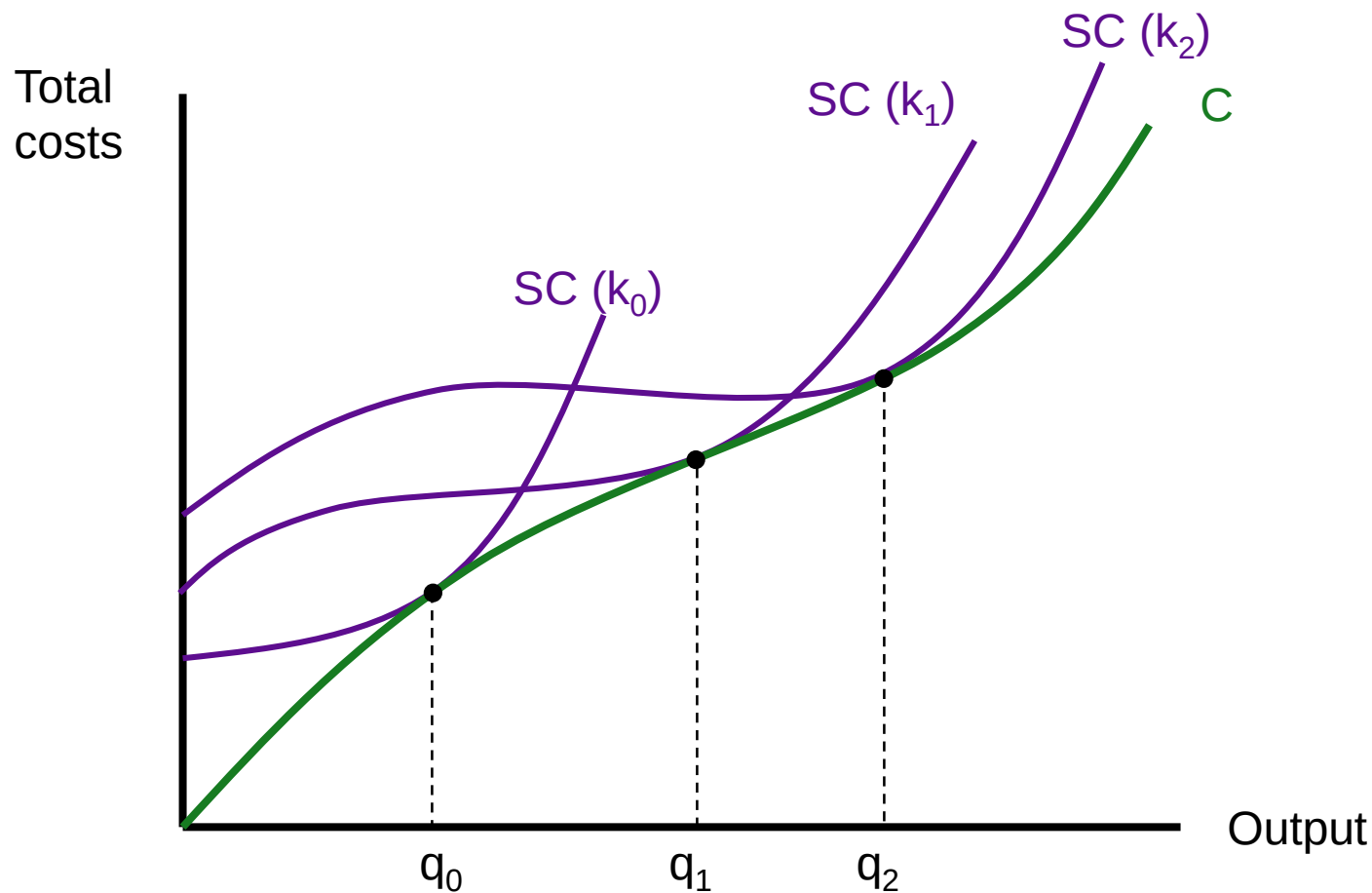
Two Possible Shapes for Long-Run Total Cost Curves



By considering all possible levels of capital input, the long-run total cost curve (C) can be traced. In (a), the underlying production function exhibits constant returns to scale: In the long run, although not in the short run, total costs are proportional to output.

FIGURE 10.8 (b) Cubic total cost curve case

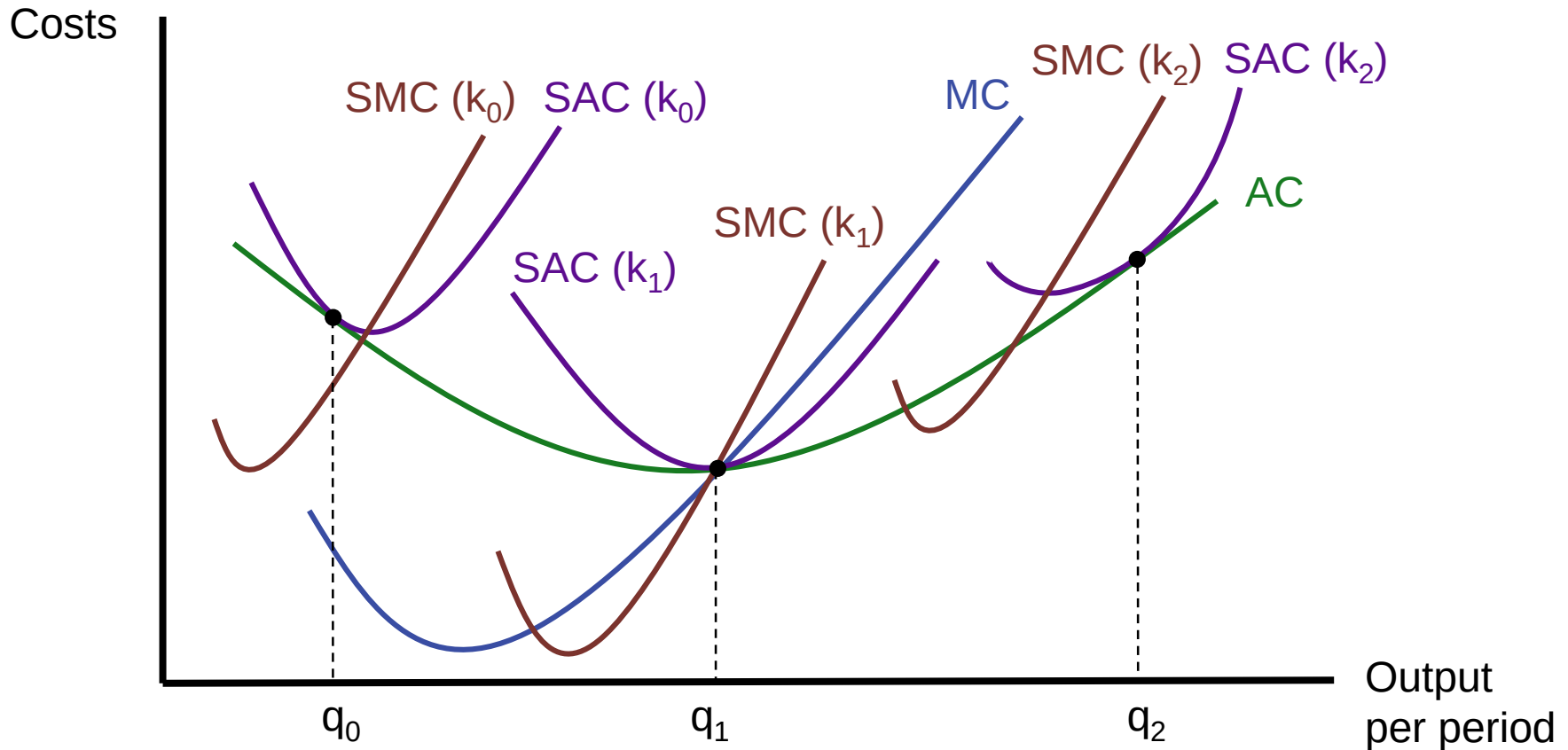
Two Possible Shapes for Long-Run Total Cost Curves



By considering all possible levels of capital input, the long-run total cost curve (C) can be traced. In (b), the long-run total cost curve has a cubic shape, as do the short-run curves. Diminishing returns set in more sharply for the short-run curves, however, because of the assumed fixed level of capital input.

FIGURE 10.9

Average and Marginal Cost Curves for the Cubic Cost Curve Case



This set of curves is derived from the total cost curves shown in Figure 10.8. The AC and MC curves have the usual U-shapes, as do the short-run curves. At q_1 , long-run average costs are minimized. The configuration of curves at this minimum point is important.

Short-Run and Long-Run Costs

- At the minimum point of the AC curve:
 - The MC curve crosses the AC curve
 - $MC = AC$ at this point
 - The SAC curve is tangent to the AC curve
 - SAC (for this level of k) is minimized at the same level of output as AC
 - SMC intersects SAC also at this point

$$AC = MC = SAC = SMC$$